

Week 2	
Objectives	The aim of this week is to understand the basic aspects of the sets of numbers that appear in Module 1, learn to correctly use their formalization, and get used to using the principle of induction for proving facts.
Session 4	Let's start with section 2 of Module 1. The MAX algorithm is a good way to understand how to deal with the induction principle. Notice how this principle applies to algorithm verification (it will be very useful to consolidate the principle). Pay special attention to the principle of induction. Try to solve self-assessment exercises 1, 2, 3 i 4. Don't do them in a hurry: it's important to write them carefully in order to grasp the concepts correctly. Do continue with complex numbers (section 3). Read carefully sections 3.1 and memorize the definition of imaginary number. We will see the binomic form of complex numbers, as well as some operations on the set of complex numbers (section 3.3). Read carefully the examples of these sections: they will definitely help you to understand the ideas. If necessary, write a summary for your study. Try to solve some of the following exercises: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 and 17.
Session 5	We will finish today with the first Module: read carefully sections 3.4, 3.5 and 3.6. In these sections we will learn the polar form of complex numbers, as well as changing from polar form to binomial form. Memorize the exponential form of an imaginary number. Read carefully how to compute roots of unity in section 3.6. It is particularly interesting that you try to re-do the examples that you'll find in these sections. It is not enough just to read them. Try to understand every single detail. Solve the rest of self-assessment exercises: from 18 until the last one.
Session 6	Use this session to solve the Challenge 1 questionnaire. Remember that the questionnaires are a learning tool, but they're also part of the evaluation process of this course, used to compute your final grade.
Keep in mind	Don't be in a hurry. It can seem that you don't have a lot of work, so use your time to be sure that you understand perfectly the concepts of this week. You'll probably have to repeat any of these sessions. Don't worry: it is an excellent investment of your time!